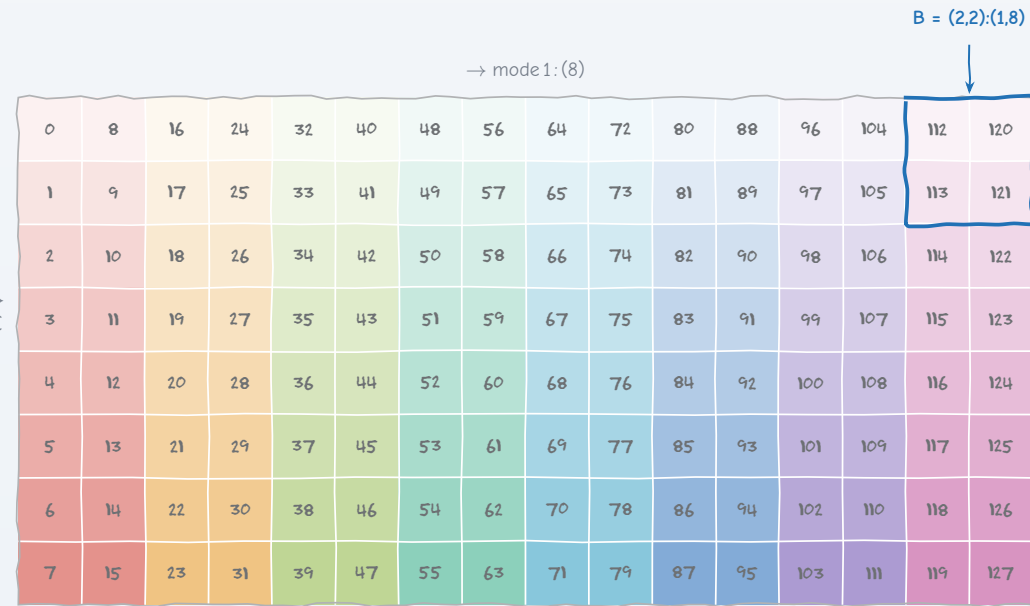


LEETCUDA

CUDA KERNEL 优化之路

从编程模型到 CUTE 与 FLASHATTENTION



L = (8,16):(1,8)

```
auto tiled_mma = make_tiled_mma(SM80_16x8x16_F32F16F16F32_TN{}, Layout<Shape<2,2,1>>{});
```

郑重声明
本书仅用于开源学习与技术交流，
未经作者本人同意，不得用于任何商业用途。

LEETCUDA PROJECT (OWN BY DEFTRUTH) · 2026 年 9 月

FA2 / FA3 ATTENTION VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) B=1, H=32, N=16384, D=128

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	232.4	1.00X	

FA2 MMA

SK=2, F16ACC	157.9	0.68X	
SK=1, F32ACC	145.6	0.63X	
SK=2, F32ACC	166.5	0.72X	

FA2 CUTE MMA

SK=1, F32ACC	189.6	0.82X	
SK=2, F32ACC	197.0	0.85X	

FA2 TMA WS (1 WG)

SK=1, SV=1, F16ACC	161.2	0.69X	
SK=2, SV=1, F16ACC	189.4	0.81X	
SK=2, SV=2, F16ACC	190.5	0.82X	
SK=2, SV=1, F32ACC	204.9	0.88X	

FA3 TMA WS (2 WG)

SK=1, SV=1, F16ACC	210.1	0.90X	
SK=1, SV=1, F32ACC	210.8	0.91X	

FA2 CUTE TMA WS (1 WG)

SK=2, SV=1, F32ACC	220.0	0.95X	
SK=3, SV=1, F32ACC	223.4	0.96X	

FA2 CUTE TMA WS PERSISTENT-CTA (1 WG)

D=128	242.5	1.04X	
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HGEMM CUTE SWIZZLE VS CUBLAS

NVIDIA RTX PRO 5000 (SM-120A) M=N=K=4096, F16ACC; FP8 4096/8192

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUBLAS BASELINE	236.9	1.00X	

HGEMM CUTE SWIZZLE (F16ACC)

S=2, BLK-SW=0	229.5	0.97X	
S=2, BLK-SW=1	231.1	0.98X	
S=3, BLK-SW=0	245.6	1.04X	
S=3, BLK-SW=1	246.7	1.04X	

FP8 CUTE GEMM (F32ACC)

WS 52 128X256, 4096 ³	443.7	1.87X	
WS 52 128X256, 8192 ³	487.8	2.06X	

SPEEDUP IS RELATIVE TO THE VENDOR BASELINE
(CUBLAS FOR HGEMM, CUDNN SDPA FOR ATTENTION).

FA SPLIT-D VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) B=1, H=32, N=8192, D=320

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	69.8	1.00X	

SPLIT-D TMA WS

SK=1, SV=1	86.6	1.24X	
SK=2, SV=2	139.9	2.01X	

SPLIT-D TMA NON-WS

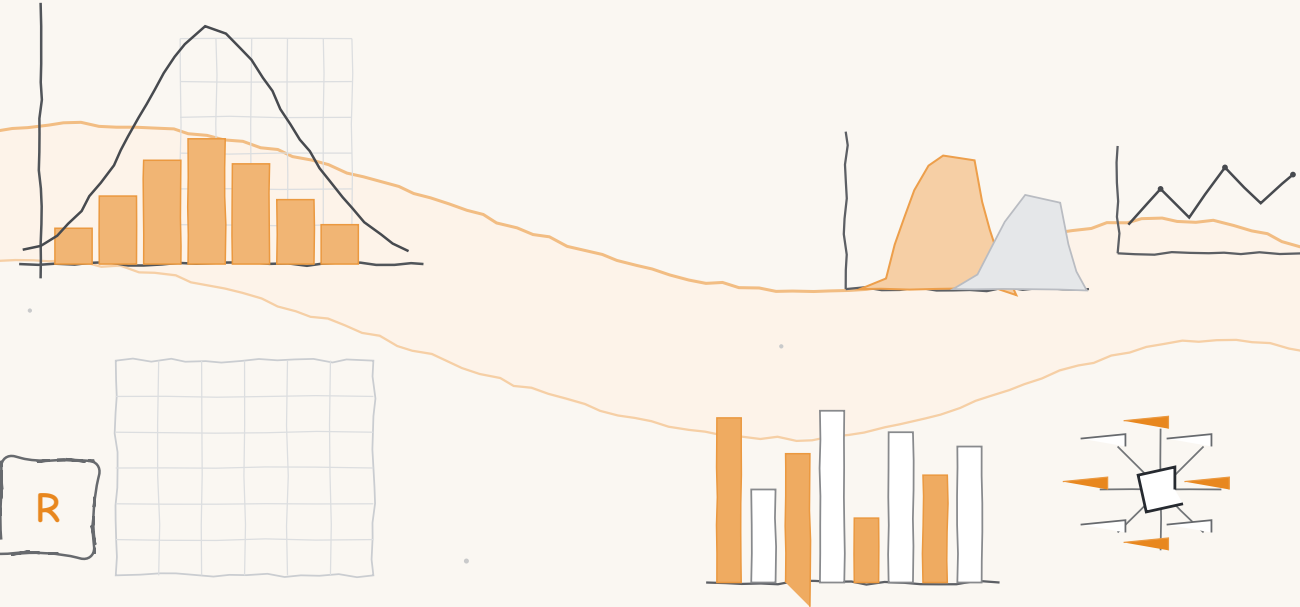
SK=2, SV=2	187.9	2.69X	
SK=2, SV=3	188.8	2.71X	
SK=3, SV=2	204.3	2.93X	

SK = K STAGES (FA), SV = V STAGES (FA), S = STAGES (HGEMM)

李航《统计学习方法》笔记

从原理到实现：基于 R

$$P(y | x) = \frac{P(x | y) P(y)}{P(x)}$$
$$\hat{y} = \arg \max_y P(y | x)$$
$$R_{\text{emp}}(f) = \frac{1}{N} \sum_{i=1}^N L(y_i, f(x_i))$$
$$H(P) = - \sum_i p_i \log p_i$$



$$p(x | \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$
$$L(y, z) = \max(0, 1 - yz)$$

$$w \leftarrow w - \eta \nabla L(w)$$
$$H(p, q) = - \sum_i p_i \log q_i$$

作者：DefTruth

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